From Concept to Submission in 8 Hours
An Agentic Research Workflow

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Building the Infrastructure

- **OpenClaw setup:** Configured an AI agent as my research assistant, connected via Telegram
- **Local models (LM Studio):** Tool calling was unreliable — had to pivot
- **DeepSeek API:** The breakthrough — cheap (~$0.15/M input), fast, reliable
- **Qwen via Ollama:** Second model for cross-checking (different architecture)
- **Whisper:** Speech-to-text for dictating ideas while pacing the kitchen
- **Dependencies:** R + glmnet/furrr, Python + matplotlib/whisper, Pandoc + LaTeX

Lesson: Start with the API that works, optimize later.

PROCOVA's Blind Spot

- **PROCOVA** is EMA-qualified for improving trial efficiency via prognostic scores
- But it assumes the **external score is perfectly calibrated** for the trial population
- **Population shift** is the rule, not the exception — especially in oncology
- **No existing method** diagnoses or corrects miscalibration using blinded trial data

Ridge-Cal

Treat the external score as a **pre-trained model** and apply a **regularized correction** on blinded data:

- **Diagnose:** Compare C-index of score alone vs. score + calibration covariates
- **Calibrate:** Ridge-penalized Cox regression on blinded data, λ by CV
- **Analyze:** Standard Cox PH with calibrated score + robust sandwich SE

6 parameters, blinded data, no additional data collection needed.

May 17–18, 2026

Time	Milestone
~21:00	Concept brainstorm (TMLE + Bayesian MCMC — too complex, scrapped)
~22:00	Pivot to ridge regression on 5 covariates
~23:00	First simulation — agent found a bug (non-PH assumption)
~00:00	10K-rep simulation running in background
~01:00	MAP-Cox bug surfaced (k applied after pooling)
~02:00	Bug fixed, re-simulation triggered

The Timeline (cont.)

Time	Milestone
~06:00	Manuscript drafted — agent wrote prose, I reviewed every section
~07:00	Sent to Gemini — Major Revision, 4 issues
~08:00	Sent to Qwen for independent review — 7 MORE issues
~09:00	Digital twin landscape report drafted
~10:00	JBS formatting, self-verify, corrections
~14:00	Voice note → small strata investigation
~15:00	Workflow codified into reusable skill

What Came Out of It

Deliverable	Format	Audience
Ridge-Cal manuscript	PDF, 12 pages	JBS submission
Response to Reviewer 2	Markdown	Journal
Digital twin survey	PDF, 12 pages	BLT (BARDS Leadership Te
Small strata white paper	Markdown	Internal stat leads
Research Workflow skill	SKILL.md	Future projects
Simulation code	GitHub (public)	Reproducibility

Why Not One AI?

Same-model reviewers share the same **blind spots**.

Internal Loop (my agent, ~$0.15/M tokens)

- Primary: DeepSeek v4 Flash | Reviewer: Qwen
- Caught: Section 4 redundancy, δ justification, missing data, table headers

External Loop (separate company)

- Gemini Pro — completely different training
- Caught: Non-collapsibility, tone, event rates, LoRA framing

Review Results

Round	Reviewer	Verdict	Issues
1	Gemini	Major Revision	4
2	Gemini	**Accept**	–
3	Qwen	Major Revision	**7 (different)**
4	Qwen	Minor Revision	–
5	Qwen	**Accept**	–

Qwen caught things Gemini missed: redundancy, δ justification, table headers, MAP-Cox framing, sandwich variance scope, missing data, references.

Takeaway: Model-diverse review converges faster than any single reviewer.

10 Minutes Start to Finish

Input: Voice note while pacing the kitchen (Whisper transcription)
"Do we need to pool small strata for CMH and MN methods?"

Step	Time	How
Transcription	<1 min	Whisper
Problem scoping	1 min	I dictated, agent structured
Independent review	1 min	Qwen caught framing weakness
Revision	1 min	I directed, agent refined
Code + simulation	3 min	Agent drafted code, 5K reps
Code review v1	1 min	Qwen caught 3 bugs
Code review v2	1 min	Verified fixes
Final + push	1 min	White paper to GitHub

Three Rounds

Review	Verdict	What It Caught
v1	Minor Issues	Scenario 4 duplicated, CMH RR variance unstratified, Wald mislabeled as MN
v2	Minor Issues	Scenario 4 label misleading, MN SE approximation
Final	Pass	Verified against bootstrap empirical variance

All bugs fixed before delivery.

Research Skill File

Phase	Description
Ph 0: Topic ID	Identify gap, map contradictory literature
Ph 1: Writeup	Write, spawn isolated reviewer, iterate
Ph 2: Simulate	2 → 20 → 200 → 10K reps
Ph 3: QC	**Code review before big runs — mandatory**
Ph 4: Full Run	Background, scheduled, monitored
Ph 4.5: Audit	Verify code matches manuscript claims
Ph 5: Manuscript	Write, verify refs, PDF self-check
Ph 6: Revision	Parse → Reconcile → Re-sim
Ph 7: Submit	Reviewer → Revise → Max 3

The Workflow: Rules

Key Lessons Codified

1. **Isolate reviewers** — fresh sessions, no cross-contamination
2. **Diversify models** — different AIs catch different bugs
3. **Code review BEFORE big simulations** — trust but verify
4. **PDF self-verify** — automated checks before reaching humans
5. **Clean up cruft** — stale files cause confusion
6. **Push to GitHub at milestones** — enables external review
7. **Batch deliveries** — pace yourself

Results

- **Independent AI reviewers caught 11 bugs** before human eyes saw the draft
- **Small tests before big runs** — avoided ~5 dead ends
- **Human-in-the-loop essential** — I directed 8 reframing cycles
- **The agent amplified my judgment, did not replace it**

Lessons Learned

- **PDF generation was painful** — 5 failed attempts with silent image failures. Now has a verification script.
- **Code review almost skipped** — hard checkpoint now in the pipeline.
- **Stale artifacts accumulated** — explicit cleanup at every milestone now.

How We Work Better Together

This morning's investigation became a case study in AI-human teamwork. Key lessons:

Use sub-agents for heavy computation. Long simulations (10K+ reps) should run in sub-agents so the AI stays available for discussion. The human should never ask "are you still there?" more than once.

Discuss before running big simulations. A 100-rep test costs seconds; a mis-specified 10K run costs 20 minutes. Check the design with the human first – they catch what no textbook will tell you.

Real-world context beats defaults. Real oncology trials use 2-look OBF at 70% IA, not 3-look at 33%. The human's experience caught this immediately.

Listen when something looks wrong. Every time the human said "that doesn't look right," there was a real bug. Running more simulations doesn't fix a wrong design.

Close the loop before scaling up. Present findings to the human

Economics

- **Session cost:** ~500K DeepSeek tokens, **well under $1.00**
- **That bought:** 1 manuscript, 1 BLT report, 1 white paper, 1 reusable skill, 3 AI reviews from 2 models, 3 GitHub repos
- **Key insight:** The agent replaces the drafting lag, not the statistician

Thank You

Repos: - `github.com/doublerobust/ridge-cal` - `github.com/doublerobust/small-strata-pooling`

The skill: SKILL.md in my OpenClaw workspace — available on request.